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Postscript on the Realism-Instrumentalism Debate

Our monograph, from its original form as an undergraduate senior essay by Seth to its present, has been in response to van Fraassen's "Perils of Perrin" paper. Hence we could not avoid having it linked to some degree to the realism-instrumentalism debate within philosophy of science. We nevertheless have endeavored throughout to avoid directly addressing, and thereby risk becoming enmeshed in, the vast philosophic literature on this topic. Ever since Part One of Duhem's *The Aim and Structure of Physical Theory*, if not before, that literature has centered on the question, *What conclusions can be drawn about posited underlying entities and processes from the explanatory successes of the theories positing them*? That question we have done all we can to avoid. Our question, again from Seth's beginning, has been far more narrow: *To what extent, if any, did the converging theory-mediated measurements heralded by Perrin of Avogadro's number and related constants gain experimental access to molecules themselves*? Van Fraassen had argued, *none at all*. We have concluded, *arguably some, yet not so much as many, including Perrin himself, have claimed*.

In particular, the experimental access to atoms and molecules gained through those measurements paled in comparison to the potential for access being gained from spectra following the Bohr model and from x-ray diffraction following the groundbreaking results of von Laue and the Braggs. From our point of view, therefore, our monograph is meant to be just what its subtitle says, a contribution to the literature on the evidential significance of converging theory-mediated measurements, not a contribution to the realism-instrumentalism literature. Hopefully, that serves as an excuse, or at least an apology, for why we have engaged that literature so little.

We shall still not be engaging that literature in this Postscript, or the question we claimed above that it has centered on. Our efforts, however, have led us to think that we can offer some useful suggestions to those engaged in the realism-instrumentalism debate. These suggestions fall into three categories covered in the three sections that follow: on the scientific literature that we found most instructive, on some distinctions we found indispensable, and, at greatest length, on the question of what kinds of scientific results have, on their face, the strongest claim to being permanent.

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P.1. On the Scientific Literature

The writings by physical scientists on the reality of molecules often tended toward polemics after the dispute over "energetics" came to a head following Georg Helm's address at the meeting in Lübeck in 1895. This tendency shows up to some extent in Perrin's *Brownian Movement and Molecular Reality* of 1909, though much more so in the various editions of Perrin's *Les atomes*, which of course was written for the general public. Another reason caution is needed in relying on Perrin's book is that the scientific community had moved beyond his values for Avogadro's number, if not already by the time its first edition appeared, by the time of its first English translation in 1916. That said, in all of his writings that we have emphasized—the October 1908 note, the 1909 monograph, the 1911 paper at the Solvay Conference, and the book—Perrin gives detailed, authoritative accounts of the experimental work done under him.

The problem with relying on Perrin is not with his accounts of his own work, but with his accounts of the measured results for molecular magnitudes published by others. We explained this in section 6.4 of Chapter 6, and hence need not repeat it here. One should always turn to the original publications to assess anything Perrin says about measurements from outside his laboratory. One point from Chapter 6 merits noting again here. Nothing surprised us more, from the time Seth first realized it, than the comparative obscurity of the Boltwood-Rutherford 1911 determination of Avogadro's number. The assumption it required to infer that value from what was actually measured, the number of α-particles that had been collected, seems to us the least adventurous of any of the assumptions required at the time to infer a value of Avogadro's number from what was actually measured—namely, that those α-particles retain their individuality when they lose their positive charge to yield helium. Moreover, of all the determinations of Avogadro's number published between 1900 and 1913, it came closest to our post-1941 value. Of course, since the experiment was not repeated, this may have been merely a fluke. It nevertheless remains an interesting question why it never attracted remotely the attention that Perrin's and Millikan's values have, both then and ever since.

The two Rutherford-Geiger papers of 1908 also merit more attention than they tend to receive for the breakthroughs they made toward the converging measurements achieved over the next few years. First, they determined a rate of α-particle emission that made possible the Boltwood-Rutherford measurement of Avogadro's number. Second, again from this rate of emission, they determined a value for the positive charge per α-particle that, far more than the determinations of *e*, had strong claim to truly being a count of the number of such charges. Third, as they expressly claimed, their implied value for *e*, the negative counterpart of half their charge per α-particle, was more to be trusted than any of the

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direct determinations of e preceding it. Their value for e from $2e = 9.3 \times 10^{-10}$, especially after it was supported by Regener the next year, became a standard against which to compare other directly determined values, even after Millikan began refining his oil-drop measurements.

Most of the attention Ostwald's *Grundriss der allgemeinen Chemie* has received within the philosophic literature has been confined to the 1909 edition and its 1912 English translation. And even then, most of the attention seems to have been centered on the three citations of Perrin, the most often quoted one from the Preface and the two scattered in the newly added Book V on Microchemistry. Assuming that we are correct that these were all inserted after the manuscript of the new edition was complete, they give a quite misleading impression of the book and the reasons Ostwald offers in it for abandoning his former position toward molecules and the kinetic theory. At least in part because so much of the manuscript had to date from before mid-1908, it precedes virtually all of the progress in microphysical measurement that was made so rapidly after it. As a result, the 1909 edition is of interest only as personal testimony about Ostwald himself and as summarizing the state of molecular science just prior to Perrin's initial publications on Brownian motion and the two Rutherford-Geiger papers.

By contrast, we regard the 1889–1890 editions of *Grundriss der allgemeinen Chemie*, both of which are available in English translation, as far more deserving of philosophers' attention than the 1909 edition. To begin with, even though Ostwald insists on the lack of evidence for the existence of atoms and molecules, his exposition is in terms of them throughout a large fraction of the book. His exposition thus displays their value in conceptualizing a large range of chemical and physical chemical phenomena, or said from another point of view, their "explanatory power." More important than this are Ostwald's frequent excursions in fine print explaining precisely why all the evidence pertaining to a set of phenomena just explicated in terms of atoms and molecules in no way presupposed them—that is, *precisely why the evidence would be the same even if the exposition had eschewed them*. We quoted some passages like this in Chapter 2, but that represented a mere sampling. As we said there, Ostwald was first and foremost an experimentalist. As such, he was able to spell out the specific respects in which nineteenth-century research in chemistry and physical chemistry had fallen short of gaining experimental access to atoms and molecules themselves, however close it sometimes appeared that it had.

The one area of research involving molecules which the 1889–1890 editions of Ostwald's *Grundriss* do not cover in any detail is the kinetic theory of gases; they simply present the theory and its account of phenomena in gases in a brief seven pages early in the book. The second, 1899 edition of Meyer's *Die kinetische Theorie der Gase*, translated that same year into English, is remarkable for the extent to which Meyer—in his own words—"collected together, as completely

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as I could, and summarized, the observations by which the admissibility of the theory might be tested and its correctness proved."[1] It is a compendium of the diverse efforts during the second half of the nineteenth century to leverage molecular-kinetic theory into means for gaining experimental access to molecules and their motions. Our citations of it in Chapter 2 do not begin to show the full range of the efforts to do this covered in the book. It was unique among all the expositions of kinetic theory well into the twentieth century in this regard—which is why others at the time, including Ostwald, Nernst, and Jeans, repeatedly cited it. Philosophers arguing about the extent to which phenomena covered by kinetic theory had provided support for the reality of molecules before 1905 have been derelict in how little attention they have given to it.

Because Meyer's book considers only phenomena covered by kinetic theory, it includes nothing on chemical phenomena and the evidence they were providing for the reality of molecules. Nernst's *Theoretische Chemie vom Standpunkte der Avogadroschen Regel und der Thermodynamik*, provides a more summary account of the evidence associated with kinetic theory in its 1893 edition, though the results it cites are in Meyer's 1899 book as well; and in the editions after 1900, Nernst cites Meyer's book as a source of more details. The special virtue of Nernst's book is that it is the one comprehensive exposition of all the evidence bearing on molecules, from chemistry and physical chemistry as well as from physics. Its first edition ran to almost 700 pages, the 1909 edition, to just short of 800, and the 1921 edition, to more than 900. Nernst as a trained physicist was always more favorably disposed to the molecular hypothesis than most chemists, yet from 1887 on he was at the cutting edge of experimental work in physical chemistry, and after Einstein's 1907 paper on specific heats, no less at the cutting edge of low-temperature experiments on specific heats—not to mention, of course, his contribution to thermodynamics, his "new heat theorem" of 1906, sometimes called "the third law" of thermodynamics.[2] Moreover, perhaps in part because his first edition was published as a complement to Ostwald's *Grundriss*, Nernst appears to us to have at all times been meticulous about when the evidence bearing on molecular theory was merely supportive of it versus when it had legitimate claim to having gained experimental access to the molecular realm.

As noted in the Appendix, Nernst's *Theoretische Chemie* appeared in 15 editions, the first in 1893, the second in 1898, and the remainder over the course of the first 26 years of the twentieth century. Five of these were translated into English, the first in 1895, followed by editions in 1904, 1911, 1916, and 1923. We consulted the first, third, and fifth of these in writing this monograph, and in the case of the 1911 and 1923 editions, we limited our attention mostly to material related to the measurements of molecular magnitudes. As far as we can tell, all editions have gone virtually unnoted in the recent philosophic literature on molecular reality. Yet, even the English translations alone

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provide the most thorough picture of how the evidence bearing on molecular reality developed from the time of van't Hoff's and Arrhenius's breakthrough results on solutions in the 1880s to the 1919 edition of Sommerfeld's *Atombau und Spektrallinien*. Moreover, no one was sociologically more in the middle of all these developments than Nernst, both before and after he instigated the first Solvay Conference.[3] Surely his *Theoretische Chemie* merits more attention than it has received from philosophers arguing about whether, when, and how the reality of molecules was established.

P.2. On Some Key Distinctions

Our monograph has relied heavily on two distinctions that are largely absent from the realism-instrumentalism debates, while trying to skirt around a third distinction (involving "approximate truth") that has been widely invoked in post-Laudan defenses of realism.[4] Here we shall explain why we think our two distinctions can make those debates more focused, even if not more readily resolvable, and why we think the third distinction is best replaced, to the same end. This will prepare the way for our thoughts on how to respond to Larry Laudan's challenge in the section that follows this one.

P.2.1. On Entering Constitutively into Research

The first distinction is between a proposition entering constitutively into further research—or, more specifically, into specific results of that research—and its not doing so. As noted in section 7.5 of Chapter 7, our distinction is at least akin to Poincaré's distinction between indifferent hypotheses and real generalizations, where, in the case of the former, results of calculations remain the same whether they are adopted or not. For us, a proposition enters constitutively into a research result just in case either the statement of or the evidential reasoning supporting the result indispensably presupposes it. Put another way, the result ceases to be warranted by the putative evidence without the proposition in question. Correlatively, a proposition does not enter constitutively into a result if it can be rejected or eliminated while leaving the result and the evidence for it intact.

Insofar as propositions can also motivate an experiment or serve heuristically in its design, careful logical analysis is often needed to determine whether any one proposition has or has not been constitutively presupposed by it. Chapter 4 has offered an example of such reasoning by showing how all of Perrin's results on Brownian motion itself did not constitutively presuppose the reality of molecules. To give just one example of why analysis was needed to conclude this,

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a proposition common to Einstein's derivation of his Brownian motion equation for ξ^2/τ and Perrin's experimental design testing it is that the motion of the granules is Gaussian insofar as Gaussian motion of molecules is producing it. Chaudesaigues, however, working under Perrin in the development of the experiment in question, took the trouble to confirm that the granule displacements are Gaussian, thereby eliminating any need to invoke molecular theory in the design of the experiment. A review of Chapter 4 will show that Perrin and his cohort similarly took the trouble to develop direct evidence, independent of molecular theory, for each proposition presupposed in the experiments yielding his Brownian motion results.

Readers might ask how we concluded in Chapter 5 that those results provided strong evidence for the kinetic theory of heat, yet not for the reality of molecules. Recall that what we concluded is that the evidence in question confirmed the existence of individual localized pressure fluctuations, on a scale notably smaller than the granules, that are in no way correlated with one another, and thus independent of one another and hence discrete. We had to invoke principles of the continuum mechanics of fluids to reach this conclusion, but not the existence of molecules. Granted that such pressure fluctuations are entirely consistent with molecular-kinetic theory and that it can therefore explain them; for that matter, it is not clear how else they may be explained. The conclusion nevertheless stands without requiring any recourse to the existence of molecules.

A handful of other examples shows how widely applicable the distinction is, beyond anything discussed in our monograph. The ether never entered constitutively into any research in optics or electromagnetism, however instrumental it was heuristically. Neither Descartes's nor Leibniz's vortices ever entered constitutively into orbital research. Leibniz's vortex theory was expressly designed to explain Keplerian orbits, but it offered no help in addressing either comet trajectories or the deviations from Keplerian motion exhibited by our Moon, Jupiter, and Saturn. It disappeared historically when research in which Newton's law of gravity was a central constitutive element handled not only all of these, but over the next century and a half so many others, in case after case revealing just what factors in the world, often quite subtle, are responsible for specific deviations from the Keplerian ideal. It entered constitutively even into the 43 arc-seconds per century residual in the precession of the perihelion of Mercury for which it failed to reveal a source, for the 43 arc-seconds are the result of subtracting from the observed 574 arc-seconds precession the 531 arc-seconds associated with the Newtonian actions on Mercury of all the other planets. In other words, Newton's law of gravity entered constitutively even into the only evidence Einstein initially had for general relativity, which helps explain why it was so important to him for Newton's law to hold in an asymptotic limit in our planetary system.

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This last example underscores the need for careful logical analysis in assessing whether a proposition is entering constitutively into further research. A good example of such analysis is Duhem's producing an entire book, expanding on his dissertation, detailing how none of the evidence for Maxwell's equations constitutively presupposed the ether.[5] Whether a proposition, as stated, enters constitutively into an experiment may become clear only at a later time. Anyone at the time would have said that the proposition, *the constituents of cathode rays are charged particles*, entered constitutively into J. J. Thomson's 1897 measurements of their mass-to-charge ratio; and anyone at the time would have taken this proposition to entail that these constituents do not exhibit wave-like properties. Yet when Thomson's son and Davisson demonstrated the diffraction of electrons three decades later, the earlier measurements were not invalidated, for what they had presupposed, in effect, was that the constituents respond in the manner of charged particles to electric and magnetic fields and upon impact with a solid. The distinction between heuristic and ineliminable constitutive elements presupposed in an experiment can be quite subtle.

The example of Newton's law entering constitutively into Einstein's evidence brings out the most important reason for insisting on the distinction. When a proposition enters constitutively into ongoing research, it will in general continue to be tested, if only tacitly and *en passant*, by that research. Moreover, as the history of research in celestial mechanics shows, this can result in increasingly stringent tests of it—as, for example, the 8 percent discrepancy *per century* in the precession of the perihelion of Mercury. We saw an example of such a stringent test in this monograph with Bohr's needing to presuppose a value for e in order to test his calculated value for the Rydberg constant against the known value. Precisely because e entered to the fifth power in the calculation, this test of Bohr's model *en passant* provided the first stringent test of the value for e, in the process historically giving the first clear evidence that Perrin's value for it, and hence for Avogadro's number as well, was not to be relied on. How stringently any experimental result tacitly, or *en passant*, tests any proposition constitutively presupposed by it requires close analysis of the design of the experiment and of the evidential reasoning associated with it. Some such indirect test is nevertheless always occurring.

Of course, ongoing research can also yield evidence for propositions not constitutively presupposed in it. The chemical phenomena that led to the introduction of structured formulas and the optical results that led to van't Hoff's stereochemical proposals added significantly to the explanatory possibilities of atoms and molecules. Those new results, however, did not test atomic or molecular theory precisely because, as we had Ostwald explain in his own words in Chapter 2, they did not require it. Van't Hoff's subsequent discovery that the constant of proportionality associated with osmotic pressure in non-electrolytic

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solutions matches the gas constant provided clear support for molecular-kinetic theory by showing that its capacity to explain phenomena of pressure in gases almost certainly extends to liquids as well. We acknowledged as much by titling the section of Chapter 2 presenting van't Hoff's results for non-electrolytic solutions "A Major Development in Support of the Molecular Hypothesis." Nevertheless, here again his research on osmotic pressure did not constitutively presuppose the molecular-kinetic theory of pressure, and hence it did not test it. At least in regard to whether these advances in chemistry and physical chemistry actually tested the atomic-molecular or molecular-kinetic hypotheses, therefore, they can legitimately be said to fall in the category of Poincaré's "indifferent hypotheses." Again, however, we stress that the logical role of any proposition in any ongoing research calls for careful analysis.

One of the central, if not *the* central, issue in the realism-instrumentalism debates has concerned what conclusions can be drawn from the explanatory successes of a theory. We do not think of ourselves in any way engaging this issue in arguing for our distinction between propositions that do and do not enter constitutively into ongoing research. The most that we are suggesting is that this distinction might add a further, hopefully clarifying dimension to some of the disputes on the issue. It brings with it, as a corollary, a distinction as well between a claim being believed without its truth or falsity making any substantive difference to the empirical content of a science and a claim, whether believed or not, that does make a substantive difference to the empirical content by virtue of research in that science constitutively presupposing it.

That molecules exist *did* enter constitutively into the research reviewed in Meyer's 1899 *Die kinetische Theorie der Gase* that was trying to use kinetic theory to obtain measured values of their sizes. The specific proposition which that research constitutively presupposed was that molecules have a characteristic dimension *s*, at least corresponding to a diameter of a sphere, that varies from one kind of molecule to another and that is a dominant factor governing the variations from one kind of gas to another of their respective deviations from the ideal gas law, their viscosities and thermal conductivities, the rates of diffusion within them, and their dielectric capacities. (We have here tried to state the proposition in the minimal form in which it was indispensable to the evidential reasoning employed in the efforts to derive values for *s* from the phenomena in question.) As Meyer acknowledged, and as we summarized in Chapter 2, the values for *s* from the five different approaches did not converge to a sufficient extent to allow anything more than an order-of-magnitude estimate for *s*, 10^{-8} cm, for molecules generally. Worse, within any one of the approaches, the values obtained for *s* for different kinds of molecules failed to meet the cross-check of yielding a single common value for Loschmidt's number—that is, the number of molecules per cubic centimeter at standard conditions—and hence too for Avogadro's number.

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The upshot of the research, repeatedly acknowledged by Meyer, was doubt, if not outright rejection, of the claim that any such *s* is physically meaningful.

Suppose, however, that the research by 1900 had yielded convergent values for *s* from at least most of the alternative approaches and that the cross-checks against the constancy of the implied values for Loschmidt's and Avogadro's numbers had been comparable to the 1912 range of ±8 percent. The conclusion would surely have been that molecules do have a characteristic dimension *s* corresponding to the diameter of a sphere—whether a dimension of the molecules themselves or, following van der Waals, of their "spheres of influence." Moreover, the values of *s* in question would have been taken as a fixed constraint in continuing research on such questions as (in Nernst's words) what "binds atoms together in molecules and hinders their flying apart." In other words, the values of *s* in question would have been seen as sufficiently well authenticated to be adopted constitutively in ongoing research on such questions. And, with that, the existence of molecules also would have acquired a standing of a very different sort than it had had before—or that it, as a matter fact, had in 1900.

P.2.2. On Taking to Be True

The standing in question is at the center of the second distinction that we think may be of some benefit in the debate between realism and instrumentalism. It is a threefold distinction among (1) a proposition, whether constitutively presupposed in ongoing research or not, continuing to have the status of a hypothesis; (2) a proposition being taken to be true and granted a presumption of inviolability in ongoing research predicated on it; and (3) a proposition that is accruing evidence and thereby becoming increasingly entrenched as a result of continuing to be tested, *en passant* yet often stringently, in conjunction with research predicated on it. The latter half of the distinction provides a response to historians and sociologists who challenge the epistemic authority of the sciences by stressing how often their most fundamental principles have been accepted on the basis of little or no evidence. The principle of inertia, for example, seems to have become a fixture well before Newton's *Principia* on the basis of nothing but Descartes's sling; but it became indirectly tested, first by Huygens's various measurements involving conical pendulums, and then by all the Newtonian determinations of celestial forces. The response generally to these historians and sociologists of science is to turn the issue of epistemic authority into one concerning the extent and stringency of the *continuing* tests, usually over decades, and sometimes centuries.

The main benefit of the latter half of the distinction for the realist-instrumentalist debate is also, we think, to shift the issue to the long-term

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evidence accruing to a "realist" assumption by virtue of the research constitutively presupposing it. We fully grant that this shift will scarcely end the disputes; it can instead focus the debate on just how indispensable "realist" claims have been to the results achieved in the continuing research.

The potential benefit for the realist-instrumentalist debate we see from the first half of the distinction is also one of refocusing. This debate often centers on what it takes to close questions like, *Do atoms and molecules exist?* once and for all—where by "once and for all" we mean the way in which perception can generally close questions like *Is there a tree in that backyard?* Sometimes the issue is whether specific evidence, or whether for that matter any evidence, can close such a question. After Worrall's "structural realism" paper in 1989[6] the issue has instead often been how an answer to such a question needs to be qualified in order for evidence to close it once and for all. The middle alternative in our threefold distinction offers a further option in these disputes: *taking* a question to be closed and predicating further research on its answer "until yet other phenomena make such propositions either more accurate or liable to exceptions." When the research predicated on it proves to be highly successful and no phenomena have yet to emerge countermanding the results of that research, a retrospective impression can be created to the effect that the question had actually been taken to be closed once and for all at the time of initial community-wide acceptance of an answer, when in truth it had then been taken to be closed only provisionally—specifically for the purpose of pursuing research along a path not available without it.

Our quoting from Newton's *Principia* is meant to remind readers that we have taken our threefold distinction from him. Our reason for emphasizing it in the body of our monograph was not because it offers a new option to those engaged in the realism-instrumentalism debate, but because it gives the correct account of the new standing the atomic-molecular and molecular-kinetic hypotheses had come to have within the scientific research community by the end of 1912. Specifically, when values for the number of molecules in a mole, their mean translational kinetic energy per degree Kelvin, and the number of unit charges required to produce a mole of monovalent gas in electrolysis had been established to sufficient precision that they were being taken for granted in ongoing research, the questions of whether molecules and unit charges exist had come to be taken as closed. While our monograph has argued for such a standing only in this specific case, we submit that its like has occurred repeatedly in the history of science.

The phrase "take to be true" alone is enough to indicate that the evidence required for a proposition to acquire this intermediate standing is something short of evidence that it is true. The point in granting it is to promote further research, yet presumably in a way that will to some extent continue to test it in

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the process. One requirement on the evidence, therefore, is that constitutively presupposing it in ongoing research will enable new evidence to be brought to bear on open questions to a greater extent than without it. The more important requirement, however, is evidence to the effect that results established in the new research which presuppose the proposition in question not be nullified should the research turn out to give reasons to revise it. The evidence must therefore give grounds at least for there being something correct about those respects in which the proposition is most indispensable to any new results predicated on it. Newton appears to have expressly rejected the viability—even the apparently unique viability—of a hypothesis's capacity to explain phenomena as adequate. The qualifying phrase he used was "propositions gathered from phenomena by induction," which on its face requires that the propositions have been shown to hold at least over a range of specific cases and hence not be purely hypothetical.

Newton's further phrasing—literally "until yet other phenomena are to have been come upon by means of which they would be rendered either more accurate or liable to exceptions"—suggests that a requirement for propositions to be taken to be true is that they be at least approximately true. The distinction invoked here, between being true and being approximately true, is the one we think is best replaced in the realism-instrumentalism debate. "Approximate truth" is an oxymoron and hence at best a metaphor. Propositions are either true or false. The conclusion Planck stated in 1913 that the fundamental unit charge is $(4.5 \pm 0.4) \times 10^{-10}$ esu was not merely approximately true.[7] Worse, as used in the realism-instrumentalism literature, "approximate truth" is much too vague to help clarify what requirements the evidence for a proposition taken to be true should meet in order to obviate the risk of results predicated on it being nullified should it subsequently turn out not to be true. What is needed in place of "approximate truth" is consideration first of the ways in which a proposition with sufficient support for it to be taken to be true can turn out not to be so, and then of provisions under which results predicated on it can nevertheless remain in place.

(Newton himself had initially adopted a threefold adverbial distinction—*exacté, accuraté*, and *quamproximé*—only to abandon *exacté* in connection with measurable quantities in the *Principia* itself.[8] He uses these adverbs to modify not the word "truth," but the entire assertion made in clauses, as in "If the areas are proportional to the times *quamproximé*, then the remaining force will tend toward the body T *quamproximé*."[9] The clause, so qualified, is itself either true or false. His main concern, in other words, was to assure that valid—that is, truth preserving—inferences could still be drawn from the antecedent to the consequent of such conditionals, even when he had decisive grounds for thinking that the empirical world deviates from it in some systematic, yet limited way.[10] Our talk of "provisions under which results predicated on it can nevertheless remain

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in place" is in this same spirit, even though the fact that the risks involved are in connection with inferences is not so apparent.)

How then can such a supported proposition turn out not to be true? One way, just as Newton said, is when subsequent results indicate that it holds to less precision than the original evidence for it revealed. Famous examples of such imprecisions are the not-yet-called "ideal" gas law, as revealed by Regnault's meticulous measurements in the 1840s, and Millikan's universally accepted 1917 value for e, $4.774 \pm 0.005 \times 10^{-10}$ esu, as revealed a decade and a half later by x-ray diffraction determinations of Avogadro's number.[11] Experimental results that presuppose any such proposition are at risk only if they depend on its having a level of precision tighter than it turns out to have. Any result not requiring it to have so much precision will remain intact, and even results requiring this can often be adjusted instead of having to discard them. An example of the latter is Huygens's measured results into the fourth significant figure for the strength of surface gravity. These presupposed uniform gravity acting along parallel lines, required as well in order to guarantee isochronism of the pendulums by means of which the measurements were made; Newton then validated the measured results by showing that they can still hold in the case of his universal gravity in an asymptotic limit as the ratio of the length of the pendulum used to the radius of the earth approaches zero.[12]

The risk of having to discard results because of limited precision can be obviated to some extent by requiring that the evidence leading to the proposition being taken to be true give clear grounds for its holding to reasonably high precision, or even better, that it provide specific bounds on precision. The marked reluctance to predicate any research on the values of the molecular magnitudes obtained prior to the 1908–1913 measurements is a sign that this requirement was being respected in practice. Keep in mind that very few in the physics community at the time had been showing any such reluctance to believe in the atomic-molecular and molecular-kinetic hypotheses.

A second way in which a proposition taken to be true can turn out not to be so is when subsequent results show that the domain over which it holds, to whatever precision, has to be markedly circumscribed. The most famous example of this is the principle Newton was insisting should be taken to be true when he enunciated his rule of reasoning, namely his law of gravity. This example, along with Maxwell's laws, was surely what Glashow had in mind when he wrote, "A physical principle cannot be said to be true until it has been shown to be false and its envelope of applicability precisely delineated."[13] As long, of course, as what new results presuppose are instances of the proposition that lie within the domain in which it holds to an appropriate level of precision, they will remain intact when that domain subsequently has to be circumscribed. The problem lies in the lack of any way to characterize, except in retrospect, the boundaries of the

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domain over which the evidence licenses the proposition to be taken to be true, and hence as well the boundaries of the domain within which results from the research predicated on it are least at risk. One way to obviate the risk is to require the evidence licensing the proposition to be taken to be true to have a wide range itself. Another is to look for complementary evidence for the results—that is, evidence for them that is less dependent on the evidence for the presupposed proposition—and ad hoc ways of cross-checking the respects in which it is dependent.[14]

Perrin's Brownian motion research included two instances of initially putting forward a conclusion, with no restriction on its "envelope of applicability," on the basis of extremely limited evidence. One was his conclusion that the mean translational kinetic energy of granules varies linearly with temperature. His evidence for this consisted of a single vertical-gradient test involving a change of temperature of less than 50 degrees Kelvin.[15] This alone gave little license to conclude, after taking the granule kinetic energy to match that of the molecules in the liquid, that α, the mean translational kinetic energy per degree Kelvin of molecules, is a constant. This was of no importance historically, for blackbody radiation measurements confirmed that $k = (2/3)\alpha$ is constant over a wide range of temperatures.

The other instance was Perrin's proposing, on the basis of the match between his measured mean translational and rotational kinetic energies of granules, "*In this, I think, will be found the most extended verification up to this day of the equipartition of the kinetic energy of translation*."[16] The quotation, in italics in the original, is from his 1909 monograph. He presents the rotational results and their match with the translational in his 1911 Solvay Conference paper, but nothing akin to the proposal of their verifying equipartition of energy. This is scarcely surprising insofar as one of the two central concerns of the Conference was the "freezing out" of degrees of freedom at low temperature and its implications for equipartition of kinetic energy.

A third way in which results predicated on a proposition taken to be true can turn out to require reconsideration is less widely discussed in the literature and hence requires a little more in the way of explanation. The proposition can come to be regarded as having been all along merely a surrogate for another proposition, usually stated in different terms, that for one reason or another is to be preferred over it—higher precision, wider range of applicability, greater compatibility with general theory, or combinations of these. To illustrate what we mean by "surrogate," consider the second law of thermodynamics.[17] It was initially formulated simply as "*Heat never flows from cold to hot*." Following Clausius's introduction of the term, this was replaced by "*The change in entropy is always greater than or equal to 0*," where *change in entropy* was taken to be an integral over a reversible process. Both of these amounted to surrogates, as we are using the term,

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for the twentieth-century statement of the law in terms of the probabilistic conception of entropy laid out by Boltzmann and augmented by Nernst's "new heat theorem."[18]

While this example illustrates what we mean by "surrogate," it does not display the risk attached to the possibility of a proposition taken to be true turning out to be merely a surrogate for a further proposition. For this we turn to the proposition that the constituent common to cathode rays, photoelectric and thermionic emission, and beta radiation—that is, the electron—is a charged particle. All the experimental determinations of the charge-to-mass ratio of these constituents, from Thomson's 7.7×10^6 abs emu in 1897 to Neumann's 1.765×10^7 in 1914,[19] presupposed them to be charged particles. So too did Bohr's model of the hydrogen atom presuppose that the electron is a charged particle, specifically a well-demarcated particle in orbit around the nucleus. None of the charge-to-mass ratio measurements had to be discarded, or even adjusted, following the conclusion that the electron is, as they say, a wave-particle, but some of the results that presupposed electrons are well-demarcated particles orbiting the nucleus did.

The original evidence on the basis of which the charged-particle hypothesis had come to be taken to be true was the response of cathode rays in magnetic and electric fields, the accumulation of charge and the increase in temperature when they impact a collector, as well as their having a measurable mass-to-charge ratio. Further evidence accrued to the proposition with each successful, increasingly precise measurement of this ratio for cathode rays and other discharges as well, especially after correction to the raw measurements introduced with the shift from *mass* to *rest mass* shortly after 1900.[20] None of this evidence, however, entailed anything about electrons in stable orbits bound to a nucleus, or for that matter, electrons anywhere except in discharges.

We intend this example to illustrate that there is a form of risk in taking a proposition to be true and predicating research on it, in addition to the risks from limitations in precision and range of applicability it may turn out to have. When the proposition turns out to be merely a surrogate for a proposition stated in different terms, some of the logical implications it had in its original statement may have to be abandoned. Any experimental results that presuppose not just the proposition taken to be true, but elements implied by the specifics of its statement, may not survive its replacement. A way of obviating this risk is to be meticulous, when predicating research on a proposition taken to be true, about which aspects of it the evidence for it specifically supports and which aspects reach beyond that evidence. Another way to express this caution is, be careful about respects in which the proposition is serving merely heuristically in contrast to its constitutively enabling a result. As we remarked in section 7.5 of Chapter 7,

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this distinction is sometimes more readily drawn in retrospect than while the research is unfolding.

These three forms of risk are not mutually exclusive. One can argue that Newton's law of gravity, for example, fell short not only in its holding to more limited precision, and that over a more highly circumscribed domain, than had been thought for two centuries, but also in its turning out, by virtue of its statement in terms of *force* akin to all other kinds of force, to be merely a surrogate for a more discriminating characterization of gravity. Needless to say, it is commonplace in the realism-instrumentalism literature to say that Newton's law is nevertheless "approximately true," and this is why so much of the research predicated on it remained in place after it gave way to general relativity.

Even if sense can be made of "approximate truth"—which we doubt—invoking it in this way is uninformative. Which results predicated on a proposition taken to be true will likely survive its replacement and which will not depends on (1) the specifics of how it enters into the different results that presuppose it *in relation to* (2) the specifics of the evidence supporting it, including both the evidence on the basis of which it was taken to be true in the first place and any further evidence that has accrued to it from its having continued to be tested, *en passant*, in the research predicated on it. *No* blanket statement of the requirements a proposition should meet before it is taken to be true for purposes of ongoing research can capture the nuances of the different ways in which such research is at risk, much less statements like "it has to be at least approximately true" or "it has to be approximately true in the right ways."

We have so far been focusing on the risks in predicating research on a proposition taken to be true when the evidence for it is limited. The most compelling reason why no blanket statement can be given of requirements a proposition should meet to obviate the risks in taking it to be true is that the risks always have to be considered in relation to the potential gains from doing so. The potential gains always concern research that would not be feasible without it. This can consist of little more than enabling a test of some novel hypothesis. An example is Bohr's taking the values of the charge and charge-to-mass ratio of the electron for granted in order to compare the value of the Rydberg constant implied by his model for hydrogen with the known value. The potential gain can also consist of being able to determine values for various quantities that otherwise were resisting measurement. Examples are Planck's contention in 1900 that taking the coefficient of temperature in his blackbody formula to be what came to be known as Boltzmann's constant allowed the values of a whole host of molecular magnitudes to be determined to an accuracy "much better than all determinations up to now," and Perrin's 1908 contention to the same effect made possible by taking the mean translational kinetic energy of granules in Brownian motion to be equal to that of molecules in the liquid.

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A still broader potential gain can consist of enabling already accessible data to be turned into much richer evidence than had been possible before. A striking example is Bohr's proposal that the lines in spectra correspond to quantum transitions from one stable energy state of an atom or molecule to another. Taking this proposal to be true enabled 80 years of spectra, which theretofore were being used at most as "fingerprints" for identifying their chemical sources, to be turned into evidence for atomic and molecular structure. The Bohr model is an especially striking example of the extent to which potential gains can outweigh the risks in taking a proposition to be true and granting it a presumption of inviolability. For, even though the experimental evidence for Bohr's revolutionary claims was initially so limited, the risks seem to have been historically outweighed by their potential for bringing an immense amount of evidence to bear on questions about atomic and molecular structure that had, until then, been purely matters for conjecture.

The Bohr model illustrates as well how research can indirectly address the risks arising from its presupposing a proposition supported by limited evidence. As we have noted, Bohr's testing his model against the Rydberg constant provided, *en passant*, a far more stringent test of the claimed values for the charge of the electron than had been possible before 1913, yielding clear evidence that Perrin's value could not be relied on, and hence neither could any of Perrin's values for the other molecular magnitudes. Finding ways for ongoing research to provide a stringent *en passant* test of propositions it is presupposing may be the most effective way to limit the range of results that might have to be discarded at some later time.

The main point of this entire section of our Postscript has concerned the virtues of recognizing an intermediate standing that a claim can have in scientific practice between that of a mere hypothesis that can explain so much and that of an entrenched principle supported by a large amount of accumulated evidence. Specifically, it can be taken to be true and employed constitutively in ongoing research on the basis of limited evidence that nonetheless shows that doing so (1) has promise of major advances in being able to develop more telling evidence and (2) has means of obviating the risks of all that research subsequently having to be discarded. We contend that a failure to recognize this intermediate standing has led to much confusion not just in the realism–anti-realism debates, but in the philosophic analysis of acceptance of claims within scientific communities generally.

Our immediate intent in this section, however, has been just to suggest a way for narrowing the gap in realism–anti-realism disputes. Realists, for example, can offer the proposed intermediate standing in response to anti-realist complaints that the only evidence for the existence of molecules as of 1913 required inference to the best explanation; and anti-realists can offer it in response to realist

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appeals to the converging values for Avogadro's number by reminding all of Nernst's words in 1921, "At present scarcely anything definite is known either of the forces which bind the atoms together in the molecule and which hinder them from flying apart in consequence of heat motion, or of their laws of action."[21]

P.3. On What Has Claim to Being Permanent

We noted in the preceding section that the value for the charge-to-mass ratio of the electron, once Lorentz's velocity correction was included, became comparatively well-defined, ranging from Kaufmann's 1.77×10^7 abs emu early in the century to Neumann's 1.765×10^7 in 1914.[22] This value changed less than 1 percent from Kaufmann's value through the advent of first the old quantum theory and then quantum mechanics and finally quantum field theory; Birge's recommended value in 1941 was $1.7592 \pm 0.0005 \times 10^7$, while the 2014 CODATA recommended value is 1.758820024×10^7.[23] One might say that, at least within tight bands of precision, this value has a clear claim to permanence. Much the same can be said for the measured values of the acceleration of gravity in Paris, from Huygens's 1673 value (in modern form) of 980.7 cm/sec^2, obtained through converging measurements from small-circular-arc and conical pendulums, to the value listed in 1958, 980.970, a change of less than 0.03 percent.[24] Huygens's value, of course, presupposed uniform gravity acting along parallel lines, not Newtonian gravity. When he put it forward he considered it a constant for the entire earth, but within a decade and a half he had concluded that the acceleration of gravity varies from one latitude to another; and we have now determined that it varies a little with time as well, in part because of seasonal and long-term redistributions of matter near the surface of the earth. Our point, however, is that the value measured at Paris has remained the same, within tight bands of precision, even though the theory of gravity has changed.

Even more notable in this regard are Ptolemy's measured values for the eccentricities and radius-ratios—that is, the ratio of the radius of his deferent to that of his epicycle—for the outer planets. Ptolemy presupposed his epicyclic model to infer values for these orbital elements from patterns in their irregular motions. Put in modern form, his values for the radius ratios inverted were 1.52, 5.22, and 9.23 for Mars, Jupiter, and Saturn, respectively, and his values for the corresponding eccentricities were 0.1, 0.046, and 0.057.[25] Kepler's 1627 values for these were 1.523, 5.20, and 9.51, taken to be the mean distances of the three in units in which the mean distance between the sun and the earth is 1.0; and 0.093, 0.047, and 0.057, taken to be the eccentricities of ellipses instead of Ptolemy's eccentric circles.[26] Our current values are 1.5236, 5.2026, and 9.5219, and 0.0934, 0.0485, and 0.0532.[27] Needless to say, both the meanings of these numbers and

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the theory presupposed in deriving them from observations changed radically from Ptolemy to Kepler, and to a lesser extent from Kepler to now. Yet the specific values obtained by Ptolemy, though more limited in precision, have more claim to correspond to current values than J. J. Thomson's original 1897 values for the mass-to-charge ratio of the electron, 0.77×10^7 and 1.17×10^7 abs emu, have.

We could give still more examples from the history of theory-mediated measurements in which the values have remained the same once allowance is made for increasing precision, yet both their interpretation and aspects of the mediating theory have changed. Obvious questions are why a theory-mediated value should ever persist in spite of theory change, what conditions have to be met for this to occur, and what significance it should be given. These questions, however, seem far removed from the realism-instrumentalism debate. So, first we should review why a claim any scientific result has to permanence has become important in this debate.

Larry Laudan's "Confutation of Convergent Realism" made this issue central to the recent debate through a series of historical examples challenging whether *any* evidence for a claim at the time can ever assure that it will not be rejected in the future.[28] This line of argument goes back at least as far as Chapter 3 of Part I of Duhem's *The Aim and Structure of Physical Theory*, where he argued that no form of experimental evidence can ever assure that claims about posited entities and processes underlying phenomena will not be discarded in the future unless the existence of the entities and processes can be confirmed by our senses.[29] Much has been written, both before and after Laudan, about the arbitrariness of Duhem's demand for confirmation by the senses. We agree with the literature on this point—indeed, throughout this monograph we have resorted to the phrase "gaining experimental access to" as providing a principled alternative for what Duhem was getting at, namely the widespread insistence within the scientific community at the time he was writing that the existence of the ether, and of atoms and molecules as well, amounted to nothing more than a hypothesis put forward in an effort to provide various phenomena with mechanical explanations. We nevertheless think that the general rejection of demands for sensible access has obscured an important element of legitimacy in the appeals that anti-realists, especially van Fraassen, continue to make to the primacy of perception. We suspect, further, that it is this element that makes questions about permanence unavoidable in the realist-instrumentalist debate.

Whatever else is to be said about perception, it does have *prima facie* claim to being able to establish some conclusions once and for all. Stated more precisely, perception *can* eliminate virtually all possibility of future information overthrowing a conclusion based on it. (The word "virtually" is included here in part to remove skeptical arguments that challenge the possibility of *any* empirical knowledge from consideration.) This is not to deny that perception can

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be misleading, or that perceptual judgments can turn out later to be false. We have so many means at our disposal for addressing such possibilities, especially collectively, that perceptual judgments nevertheless provide an exemplar of permanence in the case of empirical claims.[30]

The challenge posed by anti-realists centers on whether any claims in science can match perceptual judgments in this regard. Again put more precisely, *the challenge is whether any evidence for propositions in science can ever eliminate virtually all possibility of future information refuting them*. We submit that the charitable way to construe anti-realists' insistence on perceptual access to underlying entities and processes is to take them as invoking perceptual judgments as *the* exemplar of permanence in the empirical realm and posing the challenge of whether any propositions in science can match them. If we are right about this, arguments that perceptual access is an unfounded arbitrary requirement, however legitimate they may be in their own right, fail to come to grips with what gives lifeblood to anti-realism.

P.3.1. On Levels of Representation

With this as context, let us return to the questions we posed two paragraphs ago about theory-mediated values persisting through theory change. Central to any theory-mediated measurement is the formula relating the less accessible target quantity to the more accessible quantities in terms of which the measurement is being effected. We are calling such formulas *measures*. For purposes of illustration here we shall use two measures from Chapter 4 of the mean translational kinetic energy W of granules in Brownian motion at a given temperature:

$$\ln\left(\frac{n_0}{n}\right) = \frac{2\pi a^3\left(\rho_{gran} - \rho_{fluid}\right)g}{W}h$$

where the numerator involves the buoyant weight of the individual granules and h is the increment in the vertical distance from the plane for which the number of granules is n_0; and

$$\xi^2 = \tau\frac{2W}{9\pi a\zeta}$$

where ξ is the root-mean-square of the displacements of the granules over the time increment τ, ζ is the viscosity of the liquid, and a the radius of the granules. Notice that the only variable quantity these two measures have in common is the radius of the granules, which Perrin made every effort to assure was uniform

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across all granules in any one experiment. (We have chosen measures of W at a given temperature for purposes of illustration instead of W_T, the mean translational kinetic energy per degree of absolute temperature, because Perrin's efforts to verify that W varies linearly with temperature were so limited, and the range of temperatures feasible to test this in the case of solutions is so small.)

In an effort to avoid begging questions, we are going to analyze what enters into theory-mediated measurements in terms taken from van Fraassen's *Scientific Representation*. He would of course say that what we call a *measure* is itself a *representation*, where his point in using this term is that it resembles what it represents in some respects, but does not resemble it in others. To obtain values for the target quantity in any measurement requires observations, but he would be quick to point out that the values of the accessible quantities employed in any one determination of a value for the target quantity are rarely the observations themselves. We follow him in saying that at least two usually distinct levels of representation of the observations need to be distinguished in order to characterize the values of the accessible quantities used in a measure. The first consists of what scientists call *data*, values obtained from individual observations, but generally expressed in terms not accessible to observation. The magnitude of electric current in a given conductor, for example, is never accessible to simple observation, yet is taken as data all the time. The same is true of viscosity. In the case of the two measures we have highlighted, Perrin inferred the radii of granules in different indirect ways because he could not simply put them against a scale;[31] nor could he simply count the number of granules at any height owing to their motion, but had to resort to a series of photographs.[32]

Van Fraassen invokes Patrick Suppes's much neglected concept of a *data model* as a further level of representation of observations, or, perhaps more accurately said, a representation of the data.[33] In the case of our second measure, for instance, the data for displacements ξ have to be squared and then averaged since the measure calls for a mean value over each time interval. Similarly, both measures require a single value for the radius of the granules to be extracted from the values Perrin obtained indirectly for them in more than one way, and even with all his efforts, the uniformity of his granules is not a mere datum, but part of his data model. Suppes's original portrayal of data models was quite broad, allowing for a range of different ways "to incorporate all the information about the experiment which can be used in statistical tests of the adequacy of the theory."[34] Our appeal to the concept here is narrower: data generally have to be distilled into a form appropriate to the use they are to serve, in our case to provide values for more readily accessible quantities in a theory-mediated measurement of less accessible quantities. Data, without some form of distillation, are rarely suited for this purpose.

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A specific example of the contrast between data and a corresponding data model can clarify why the distinction is important for our purposes. The data in the one-second small-arc circular pendulum measurements of surface gravity initiated by Huygens consisted of the length of the pendulum, measured to the center of a spherical bob, and the duration over which the pendulum beat synchronously with a well-tuned pendulum clock, typically required to be of the order of a half-hour or more. Huygens called for two corrections to these observations: a calculated adjustment of the length to the center of oscillation of the pendulum, resulting in a length slightly longer than the measured length to the center of the bob; and a further correction to the length to compensate for any imprecision in the measurement of time by the clock versus sidereal time, checked over the duration of a sidereal day.[35] Newton subsequently introduced a small correction for the reduction in effective gravity owing to buoyancy of the bob in air. The data model yielding the measurement of acceleration at the surface of gravity included these corrections of the raw data. Over the course of subsequent centuries other corrections came to be included, such as one for the resistance of the air to the motion of the bob, another for the deviations from isochronism as a function of arc-length of any circular-arc pendulum.[36] This example illustrates why the distinction is so important in precision measurement: the distillation of data into a data model, among other things, can compensate for sources of systematic error that would otherwise limit the amenability to increasing precision of a measure.

While much more can be said about data and data models as levels of representation of observations, it does not have much bearing on our concern here—why results of theory-mediated measurements can persist through theory change. Our point so far is that two levels or layers of representation precede the use of a measure in obtaining values of a target quantity. The measure itself is a still further level or layer of representation in van Fraassen's sense insofar as it usually represents a relation among quantities in some, but not in other, respects. More important than this, however, is that the measure does not come out of nowhere. It requires justification first for the specific mathematical form it asserts to hold among the various quantities entering into it, and second for what it claims its target quantity amounts to physically. This is true even when the target quantity is, historically, being invented with the measure. As Hasok Chang stressed in his *Inventing Temperature*, efforts were required first to show that two specific heights of mercury in a cylindrical tube consistently correspond to the freezing and boiling points of water at atmospheric conditions, and second that the relationship between the height of mercury intermediate between these two heights had claim to varying linearly with intermediate stages between the freezing and boiling points.[37]

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Putting cases in which a quantity is being invented with a measure to one side, the justification for a measure usually consists of a derivation of it from broader mathematically expressed principles. Thus, the measurement of electric current by the magnetic torque on a coil carrying it serves to define the measure of it incorporated into galvanometers and ammeters. The specific expression relating current in the coil to the torque is derived from the more general principle specifying the force on a current carrying conductor in a magnetic field, and that principle in turn can be derived from the general mathematical theory defined by Maxwell's laws. As we summarized in Chapter 4, the two measures for *W* cited earlier were both derived, by applying the *mathematics* of kinetic theory to solutions, under assumptions about the motions of granules, several of which were independently tested. Insofar as the justification for a measure involves considerations reaching beyond it, it too must either itself be or be part of a further level or layer of representation, namely one that links variables to one another in some more general fashion. For want of a fully descriptive name here, let us call this level of representation *mathematical theory*, leaving open the extent of theory invoked in justifying a measure. We repeat that what is required of this fourth level is to justify first the specific mathematical form of the relationship among the quantities in the measure and second what the measure takes its target quantity to amount to physically.

These four levels of representation invariably enter into any theory-mediated measurement, and hence the question of why values from some of these measurements persist through theory change needs to focus on them. Duhem explicitly (and van Fraassen at least by implication) identifies a still further level of representation, the "explanatory part" of physical theories that he then goes on to license in the form of heuristic "as-if" models.[38] What he had in mind were the ether models of electromagnetic phenomena put forward by Maxwell and his followers, but surely too Ostwald's 1890 concessions to the conceptual virtues of atoms and molecules in physical chemistry.[39] Here we prefer to draw the distinction between this level of representation and the level we have called "mathematical theory" in the manner of an avowed realist, the physicist Steven Weinberg. He distinguishes between a "part of modern physical theories . . . that usually consists of the equations themselves, together with some understandings about what the symbols mean operationally and about the sorts of phenomena to which they apply" and a part that "is the vision of reality that we use to explain why the equations work."[40] Taking this last phrase as the key, let us call the fifth level of representation *explanatory theory*.

Again, a lot more can be said about the distinction between what we are calling *mathematical* and *explanatory* theory, much of it contentious because of the ways in which it ties into so much of the realism-instrumentalism debate. For our purposes here, fortunately, we need not say much more, for the main function of

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the five levels of representation that we have distinguished is to situate the middle of the five, what we have called the *measure* itself, in its broader logical context. We can now concentrate on it, asking why it can, at least in some cases, yield values of its target quantity that persist through theory change, where the latter allows for changes in both our "mathematical" and our "explanatory" theory.

P.3.2. "Stability" as a Factor in Claims to Permanence

In Chapter 7 we concluded that Perrin had provided evidence that each of his two measures for W that we are using for illustration is *stable*—that is, evidence that the value for W obtained by means of each measure remains the same to reasonably high precision over a reasonably wide range of their respective manipulable quantities. To quote Perrin in regard to the first measure, "the values of the granular energy deduced from the preceding series are concordant within the limits of accuracy of the experiments, although the mass of the granules has varied as 1:40, the difference of density between the granules and the medium has varied as 1:4.7, and the rapidity of rarefaction as a function of height has varied as 1:30."[41] A similar statement holds for the other measure, for which Perrin varied the radii of the granules from 0.212 to an extreme of 5.5 microns. Evidence for the stability of each measure is evidence as well that a specific linear relationship holds in each of the two cases: between $\ln(r/r_0)$ and h in the first case, with its slope varying with the radius of the granules and their buoyant density; and between ξ^2 and τ in the second case, with its slope varying with the radius of the granules and the viscosity of the liquid.

Now for the important point. The values of each of the manipulable quantities just listed were determined completely independently of the mathematical justification of the respective measures! Therefore, the *value* of W obtained through the measure, in and of itself, amounts to nothing more than a factor in the value of the coefficient of proportionality of each of two empirically determined linear relationships between two variables whose values in no way depend on the mathematical justification of the measure. In other words, *the empirically determined value for W and its empirically turning out, at least to reasonable precision, to be a constant over a range of the manipulable quantities are, in each case, independent of the mathematical justification of the respective measures*.

This is not to say that the value determined for W and its constancy are independent of the respective data and data models employed in the two determinations. And, of course, that W represents the mean translational kinetic energy of the granules at the given temperature in each case is entirely dependent on the mathematical justification of the respective measures. Our point is that the

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numerical value itself obtained for *W*, putting to one side what it amounts to physically, is in each of the two cases simply an empirically determined numerical factor in the coefficient of proportionality in an empirical relationship among variables; and the values of all those variables, and hence too the *value* of *W*, are determined independently of the mathematical justification of the measure. As a consequence, evidence for the stability of each of the measures of *W* is evidence that a linear relationship holds empirically among certain specific quantities over a certain range of values of the manipulable variables, and this evidence is entirely independent of the mathematical justification of the measure.

This point can be put in another way. Evidence for the stability of each of the measures of *W* is evidence for the existence of a *regularity* in the world corresponding to each of the measures, a regularity that is being *represented* as a relation among comparatively accessible quantities: in the first case, among the ratio of the number of granules in a narrow slice at different heights, those heights, the radii of the granules, and their buoyant densities; in the second case, among the mean of the square of the displacements over different time intervals, the magnitudes of those time intervals, the radii of the granules, and the viscosity of the liquid. To complete the specification of the relationship in each case, a value for *W* is needed. Evidence for the stability of the measure is evidence that this value of *W* is a numerical constant in the relationship, and hence a constant whose value can be determined from values of the other quantities in the relationship, themselves represented in the data model employed in the measurements. The *value* of *W* accordingly has an empirical integrity in its own right, independent of the mathematical theory underlying the measure, and hence independent too of what it physically amounts to.

The same point can be made for Huygens's theory-mediated measures of the strength of surface gravity and Ptolemy's theory-mediated measures of the radius ratios and orbital eccentricities of the outer planets. The value of Huygens's distance of fall in the first second in the absence of air resistance, half our *g*, taken unto itself, was a factor in a coefficient of proportionality of a linear relationship between the length of pendulums and the square of their periods. The regularity represented in the case of his simple pendulum measure, which was restricted to cycloidal and small-arc circular pendulums, was between two quantities that could be determined independently of any assumptions about gravity, and hence so too could the value itself for the strength of surface gravity. (The same is true of Huygens's conical pendulum measures, of which he had two, the first restricted to constant-height pendulums and the second to ones in which the center of the bob is restricted to the surface of a paraboloid; both involved a regularity represented as a relationship between the square of the period and, in the case of the former, the height of the pendulum and, in the case of the latter, the latus rectum of the paraboloid.[42])

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In saying that the measures represent a regularity, we do not mean that first there was the regularity and then came its representation by the measure. That such a regularity exists, in both the Perrin and the Huygens cases, was initially derived from theoretical considerations. These derivations, however, could not guarantee that the regularity actually exists. That it does exist gets confirmed by what we are calling the stability of the measure—a conclusion that is reached by means of inductive generalization from cases. One can legitimately take its empirically established stability as evidence supporting its derivation and hence the theoretical claim that it exists. Much of our Chapter 4 was devoted to spelling out just what the evidence of this sort consisted of in the case of Perrin's two measures. Here we are making a different point. Once the stability of the measure confirms the existence of a regularity among its accessible variables, *the existence of this regularity no longer depends on the theoretical derivation of the measure*. What that derivation then provides is a physical interpretation of the less accessible quantity that is being measured. This explains how the value of that quantity itself, though of course not its physical interpretation, can gain an integrity in its own right, independently of the theoretical considerations underlying the measure.

The regularities we are singling out here are instances of the "gross regularities" that Jody Azzouni has stressed in various places, most notably in his *Knowledge and Reference in Empirical Science*.[43] Two distinguishing features of such regularities, according to him, are their highly local character and their existence being amenable to confirmation independently of theory and hence their withstanding theory change. He has argued that they are not only ubiquitous, but that they are far more crucial to understanding science than has generally been recognized.

Our point here is more modest, yet nonetheless again more crucial to understanding measurement in science than has generally been recognized. Most theory-mediated measurements involve a comparatively local regularity, the existence of which and hence the utility of which for purposes of measurement are independent of overarching theory, thereby tending to insulate them from theory change. This is true not just for the examples we are considering here, but for ones establishing quantities in science in the first place, like the regularity between the boiling point and freezing point of water, on the one hand, and the length of a column of mercury, on the other, or the correlation Ampère discovered between the number of voltaic cells to which a wire is connected and the response of a magnetic compass adjacent to that wire. Our main point here is narrow, namely that the stability of a measure typically provides evidence for both the existence of the regularity at its heart and its independence of theory.

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Because Ptolemy's measures are expressed in a geometric form, the corresponding points are more hidden in their case. Nevertheless, his values for the radius ratios and eccentricities of Mars, Jupiter, and Saturn were part of a representation of regularities in their respective observed departures from uniform angular motion; and the values themselves were determined from these departures, and thus had an integrity of their own, independent of his epicycle model.[44]

In none of these cases are we denying that the target quantities of the measures are being taken to represent physical quantities. Our claim is that their physical interpretation does not come from the data or data model used in determining their values. The physical interpretation in every case lies in the mathematical justification of the measure—which, in our terminology, is part of the mathematical theory underlying it. This terminology is misleading in one respect, however. The measures are not simply being deduced from a mathematical theory, established or otherwise. Their mathematical justification invariably involves assumptions. The key assumption in the case of Huygens was that gravity acts along parallel lines to yield uniform vertical acceleration. In the case of our second illustrative example from Brownian motion, the assumption was more complex: (1) the granule displacements over time are random, with a Gaussian distribution over time; (2) the resistance force from the viscosity of the liquid on the spherical granules is in accord with Stokes's law; and (3) the duration τ covers a very large number of changes in velocity. Perrin took steps to confirm the first two of these independently of the measure, in the process yielding a historically landmark conclusion in the case of the first. Nevertheless, the claim that W represents the mean translational kinetic energy of the granules depended on these assumptions, and thus had a hypothetical status that the value obtained for W, in and of itself, did not have.

This stands out even more clearly with our other illustrative measure from Brownian motion, for it required an assumption about the relationship between the pressure exerted on the walls by impact of the granules and the fraction of the vertical pressure increase in the fluid from the weight of the granules, neither of which could be independently measured.[45]

To summarize, while the value determined for the target quantity in each of these cases depended only on its data and data model, not on the mathematical justification of the measure, the physical interpretation of the target quantity depended indispensably on the latter, and hence had a hypothetical status owing to the status of the assumptions made as part of the justification. The numerical value Perrin obtained for W in each of our two cases was accordingly not open to challenge in the same way as his claim that this value is the mean translational kinetic energy of the granules in Brownian motion. Among the challenges at the time was whether Stokes's law holds for granules so small.[46]

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P.3.3. "Convergence" as a Factor in Claims to Permanence

We have so far been considering our two illustrative measures of W separately from one another. We now consider them together. At the time of the Solvay Conference in 1911, Perrin's values for them at the same temperature agreed with one another to within 0.6 percent—far closer than the roughly 10 percent and greater gap between his values for W and the values for the mean translational kinetic energy of molecules at the corresponding temperatures implied by the measurements of e in the laboratories of Rutherford and Millikan. To put the point in the way we have in the main body of this monograph, Perrin's two stable measures of W had far stronger claim in 1911 to having yielded *convergent* values than the various stable measures of N_0, e, and α had. The question now is, what further conclusions, beyond those we have drawn from the stability of the two measures, are to be drawn from their convergence?

The first point to notice is that the convergence was of no consequence at all for the specific *values* obtained from the two measures. The values themselves were obtained as factors in the coefficient of proportionality in the linear relationships between, respectively, $\ln(n/n_0)$ and h, on the one hand, and ξ^2 and τ, on the other. Each value for W depended only on the data and data model from which the values of all the other quantities in the measure were drawn. That the two values agree with one another, however, does provide evidence that neither one of them, nor the stability of the respective measures, was merely an artifact of the data, data model, or assumptions underlying the measure. Absent the convergence, the claim that neither value is merely such an artifact relied only on the mathematical justification of the measure. Evidence that the values are not artifacts is evidence that they are values representing a physical magnitude. Moreover, insofar as a lack of convergence would have been evidence that W does not stand for the same quantity in the two cases, the convergence was evidence that the symbol does stand for the same physical quantity in both measures.

Nothing in the data, data models, or measures, considered in isolation from the theoretical justifications of the latter, assured that W stands for the same quantity in both. Indeed, the only other term common to both is the radius of the granules, a, and it enters as a cube in one of them and linearly in the other. Why these two measures should have a factor in their respective coefficients of proportionality in common can therefore be answered only by turning to their respective mathematical justifications. The mean translational kinetic energy of the granules enters into the two justifications in very different ways—in the first case in connection with the pressure gradient needed to support the weight of the granules; and in the second in connection with their motion resisted by viscous forces.[47] The convergence of the two measures was accordingly evidence that two distinct regularities in Brownian motion, represented by the respective

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measures, have a governing physical magnitude in common, represented in each measure as the mean translational kinetic energy of the granules at the given temperature. But then the combination of the stability of each measure and their convergence was evidence in support of the assumptions made in the theoretical derivation of each of them—not so much that these assumptions are true, but that they were safe to adopt for purposes of measurement.

We have expressed this point in such a cautious fashion in an effort not to beg any questions in the realism-instrumentalism debate. The point can be made in a less roundabout way. Before turning to their convergence, we acknowledged that the claim of each to be a measure of the mean translational kinetic energy of the granules at the given temperature was open to challenge because of assumptions made in its mathematical justification. The stability of each measure provided evidence that its *W* was a constant at any given temperature over a range of globule sizes and liquids—a remarkable finding in itself—but not grounds for dismissing challenges to the claim of each that what it measures physically is the mean translational kinetic energy of the granules. The convergence of the two measures, by contrast, did give grounds for dismissing such challenges, for it provided evidence that these two measures are measures of the same physical quantity, with each measure holding to a certain level of precision over a certain range of values of its manipulable variables. In other words, their convergence provided evidence that any shortcomings in the assumptions entering into the justification of each measure—or, for that matter, in the respective data models—were at worst imposing limits on its precision, on the range over which it holds, or on some combination of these two.

In the preceding section of this Postscript we spoke of one proposition coming to be regarded as all along having been merely a surrogate for another proposition that for one reason or another replaces it—higher precision, wider range of applicability, greater compatibility with general theory, or combinations of these. Insofar as the factors listed at the end of the last sentence of the preceding paragraph parallel these, we can restate its point: *the combination of the stability of the two measures and their convergence provided evidence that each of them would turn out to be no worse than a surrogate for some other measure that has higher precision, a wider range of applicability, greater compatibility with advances made in mathematical and explanatory theory, or some combination of these*.[48]

To conclude that the evidence showed each measure to be at least a surrogate for some more preferable one is equally to conclude that this evidence is imposing a constraint on replacing either measure with any alternative to it: any replacement must either retain the measure as holding to a certain level of precision over the range of the manipulable variables for which it was stable and converged with the other one; or it must explain why the evidence arising from their stability and convergence was somehow illusory. Putting aside the second of

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these alternatives, no new measure can meet the first demand unless it preserves, at least to high approximation, the *values* for *W* that had been obtained by means of the original measures. We are thus beginning to see why the values themselves obtained through theory-mediated measurements *can*, at least in some cases, persist through theory change. The values, when supported by sufficient evidence, tend themselves to become constraints on any change in theory in which the original measures are replaced.

The examples we have been using as illustrations are historically of no interest when it comes to theory change, for the values obtained for *W* ceased being of much consequence to physics once it became clear—surely no later than 1915—that they are 10 percent (or more) less than the mean translational kinetic energy of molecules at the same temperatures. A more historically illustrative example involving theory change is Huygens's converging measures of the distance of fall in the first second in the absence of air resistance. Both his simple and his conical pendulum measures presupposed, in modern terms, that the acceleration of gravity does not vary with any varying heights of the pendulum during the measurements, and that it is vertical, along strictly parallel lines. Both measures proved to be highly stable, and in his own statement of the results in 1673, the values from the two agreed to better than three significant figures. Both measures therefore amounted to mere surrogates in the context of Newtonian gravity—including the replacement of *g* taken to be a constant vertically with *g* varying with height. To use Glashow's phrasing, in showing Huygens's measures to be false, Newton also delineated their envelope of applicability, thereby validating the value Huygens's measures had obtained for *g* to the better than three significant figures claimed for them. Newton then invoked this value in his evidence that terrestrial gravity extends to our moon.[49] In this case, therefore, the value was more than just a constraint on theory change; it was crucial to the evidence for extending terrestrial gravity to the celestial realm.

P.3.4. "Amenability to Increasing Precision" as a Factor in Claims to Permanence

In Chapter 4 we noted a further demand that is customarily made of theory-mediated measures: their amenability to increasing precision. A classic example was Huygens's redefining the length of the pendulum once he had solved the center of oscillation problem.[50] Perrin himself, in one of his first papers on Brownian motion, had claimed that the first of his measures for *W* that we have been considering "was susceptible to an unlimited precision."[51] We also noted in Chapter 4 that Perrin, in a later edition of *Les atomes*, had said that René

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Constantin had convinced him that the motion of granules near the walls was too constrained to be representative and had then added:

> Working at a sufficient distance from the walls with the grains I had used, he obtained the value $N = 64 \times 10^{22}$; unfortunately the number of observations (about 100) was too small. These measurements will be repeated.[52]

This change concerned the second of our two measures, amounting to a 7.5 percent increase in the value of *W*. Notice that the pursuit of higher precision here concerned the data and data model employed, not the measure itself, though what prompted Constantin to propose it was the conflict between the assumption of unbounded fluid in Einstein's derivation of his relationship between ξ^2 and τ, on the one hand, and the contained emulsion, on the other. What difference might have emerged in the value for *W* from the second measure had Perrin followed up this proposal we have no way of knowing. We should nevertheless ask, while we are at it, what contribution the amenability to increasing precision can make to the persistence of values from theory-mediated measures across theory change.

To begin with, evidence that a measure is amenable to being made more precise is evidence that it is measuring a definite physical quantity. Planck illustrates this point in the 1913 edition when he reformulates his derivation of the molecular magnitudes from phenomena of radiation in terms of two more highly measurable constants, one, *a*, from the Stefan-Boltzmann relation and the other, c_2, from the "specific intensity of a monochromatic plane polarized ray of the wave length λ which is emitted from a black body at the temperature *T* into a vacuum perpendicular to the surface."[53] He then derives values for his *h* and the various molecular magnitudes to three significant figures, ending with the claim: "Since absolute accuracy is claimed for the formulae here employed, the degree of approximation to which these numbers represent the corresponding physical constants depends only on the accuracy of the measurements of the two radiation constants *a* and c_2."[54] In other words, if the formulas are exact, the amenability to increasing precision of the various measures depends only on the data and data model for the two radiation constants. Equally, however, if increasing the precision of the measurements of the two radiation constants were to result in a more precise value for any of the molecular magnitudes in conflict with a value of it from any other measure, the question would arise whether Planck's formulas are "absolutely accurate."

Amenability to increasing precision of a measure thus opens the way to assessing the limits of precision of alternative convergent measures of the same quantity against one another and deciding which one is to be preferred. To put this in a more customary manner, amenability to increasing precision in the case

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of stable, convergent measures becomes a research tool for exposing sources of systematic error in the measures. This is precisely what happened in the case of Millikan's 1917 value for e, $4.774 \pm 0.005 \times 10^{-10}$ esu. It was being taken as the preferred basis for all other molecular magnitudes until the 1930s, when comparison with its implied value for Avogadro's number with a value obtained from x-ray crystallography exposed a roughly 0.7 percent systematic error in it, introduced by a mistaken value for viscosity of air. This error in the data model was corrected to yield an oil-drop value for $e = 4.8130 \pm 0.0006 \times 10^{-10}$ esu; in the process, however, it was decided that limits of precision in the measurement of viscosity entailed limits to the amenability of increasing precision in the oil-drop measure, and this measure then ceased being taken as the preferred basis for all other molecular magnitudes.[55]

The upshot, then, is that amenability to increasing precision amounts to a way of cross-checking values obtained from stable, convergent measures, a way of addressing the risk of being misled by a value that appears to have greater precision from its determination than it should be taken to have. The greater the precision that a value can come to claim, the greater the reason to assure its persistence, with suitable bounds on its claim to precision, through any subsequent changes in theory. Indeed, what actually persists historically, as the Millikan example shows, is not the original value, nor even one subsequent value, but the increasingly precise variants of them. That still leaves the original value as an approximation to subsequent values, and the fact that precision unfolded in a sequence from the original amounts to evidence not just that the original value itself held to a certain limited level of precision, but that the sequence of more precise values is at least approximating, to an increasing level of precision, the value of a definite physical quantity.

The history of our two illustrative measures for W brings out another way in which the demands for convergence and amenability to increasing precision can lead to reconsideration of a measure. We have presented them as measures of the mean translational kinetic energy of granules. Both Einstein and Perrin assumed that W is at the same time a measure of the mean translational kinetic energy of molecules in the solution at the same temperature, licensing the two of them to transform both of these measures into measures of Avogadro's number. And such they were viewed throughout the period covered in this monograph. Their values for this constant, however, remained 10 percent (and more) greater than any of the values that were being obtained from the measures based on gaseous ionization, α-particles, and blackbody radiation, all of which were within 3 percent of one another. As Millikan's value for the first became increasingly precise, the Brownian motion measures of Avogadro's number came to be considered as, at best, merely approximate.[56]

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Insofar as the Brownian motion measures converged with one another, this implied either that they contained a systematic error or that *W* should be taken to be a merely rough, approximate measure of the mean translational kinetic energy of molecules. But the two convergent measures of *W* we have been considering had no parameter in the data from which they were obtained in common. For their deviation from other measures of Avogadro's number to be coming from systematic error therefore required two separate systematic errors of nearly matching magnitude. Far more likely was that the mean translational kinetic energy of Perrin's granules was at least 10 percent less than that of the molecules in his emulsions and therefore was a quantity that only correlated with that of the molecules, not a measure of it amenable to increasing precision. We do not know of anyone who expressly made this point at the time, but as interest in the Brownian motion measures of Avogadro's number diminished, no one may have had occasion to ask just what Perrin's measures were in fact measuring. Notice, nonetheless, that this in no way undercut the evidence from their stability and convergence that they were yielding reliable values for the mean translational kinetic energy of his granules.

P.3.5. Claims to Permanence

The point of both this and the Millikan example is that the demands for stability, convergence, and amenability to increasing precision can provide very strong cross-checks on claims made for the value yielded by a measure. At the beginning of this section we proposed that in the case of perception, one *can* eliminate virtually all possibility of future information overthrowing a conclusion based on it, and in that regard anti-realists have some grounds for insisting on it as a standard against which claims to permanence in science should be compared. Our examination of why certain values from theory-mediated measurement have persisted through theory change suggests an at least analogous point for them: *the demands for stability of a measure, its convergence with other measures of nominally the same quantity, and its amenability to increasing precision* **can** *eliminate virtually all possibility of future information showing that the* **value** *itself obtained from it was all along just an artifact of the data or data model from which the value was extracted, and therefore all along not something corresponding, at least to a degree of approximation, to a bona fide physical quantity.*

The emphasis on the word "can" is to acknowledge that this does not happen automatically; it depends on the strength of the evidence for stability, convergence, and amenability to increasing precision, and perhaps as well to specifics about the measure. The emphasis on the word "value" is to distinguish it from its physical interpretation, for we do not want to be suggesting that the physical

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interpretation of the value can have a comparable claim to permanence. We have seen earlier with Perrin, Huygens, and Ptolemy that a value obtained from theory-mediated measurement can have a claim to permanence, and with it a claim to being at least an approximate value for a physical quantity, without its physical interpretation becoming closed to reconsideration.

How the value can have such a claim to permanence stems from two aspects of theory-mediated measurement. First, the whole point of such measurement is to determine the values of less accessible quantities from those of more accessible ones. It does this by finding a regularity among the more accessible quantities—or, expressed more in keeping with van Fraassen, a representation of a regularity expressed in terms of the more accessible quantities—in which the target quantity is a parameter. Examples of such regularities are the period of a small-arc circular pendulum and its length and the rate of rarefaction with height of granules in a dilute emulsion and their buoyant weight. The target quantity is required to have a specific value in any one instance of the regularity, but it need not be a constant: *g* turned out to vary from one location to another, *W* was all along expected to vary with temperature, and Ptolemy's radius ratio and eccentricity varied from one outer planet to another. The regularity, moreover, need not be some general law of nature; it need hold only under a quite restricted, localized set of conditions. For determining values for *W,* Perrin's rarefaction regularity was required to hold only over a limited range of heights for dilute emulsions formed with uniform spherical granules of a limited range of sizes—scarcely a fundamental law of nature, notwithstanding the fact that Perrin's Nobel Prize citation included the phrase, "especially for his discovery of sedimentation equilibrium."

Second, insofar as the measures require only relatively localized regularities among comparatively accessible quantities, they are subject to extensive testing and hence numerous cross-checks. Evidence showing that a measure meets the demands for stability and for amenability to increasing precision requires just such testing. The aim with both demands is evidence that the measure represents, at least to a certain level of approximation over a certain range of its accessible quantities, a physical regularity, and hence that the value it yields is not an illusory artifact of the data and how they are modeled.

The key point here is that what is being tested is a regularity within the accessible quantities, and whether this regularity exists in fact is independent of the theoretical justification of the measure, and hence of the physical interpretation of its target quantity. The demand for convergence with another measure of nominally the same quantity seems primarily to be a test of its physical interpretation. Even so, an agreement between two numerical values obtained from different regularities holding among different accessible quantities is evidence that in neither case is the value itself somehow merely an artifact of how values are being determined for the respective accessible quantities. At least in principle,

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therefore, these three demands can address and eliminate a great many ways in which future developments might lead to the conclusion that the *value* obtained for the target quantity all along failed to correspond to any physical magnitude.

The parallel we are suggesting with perception should now be clear. The eliminate-virtually-all-possibility claim in the case of perception is based first on the many resources we have for addressing different sources of perceptual illusion, second on the extent to which these resources depend on perceptual regularities, and third on the extent to which these regularities allow such sources, save for ones involving radical skepticism, to be addressed locally. Verifying that a perception is not illusory, moreover, can be done independently of the words used to describe its content. The process of verifying it may even lead to a reverbalization, yet with no change beyond refinement of detail in the perceptual content itself.[57]

The parallel with perception extends beyond the ways we have to eliminate sources of illusion. The theoretical justification of any measure amounts to a prediction that a certain regularity is going to be observed among the accessible quantities and that, by virtue of this regularity, the measure will yield a stable value for its target quantity. This is a prediction, however, not a guarantee. Whether the regularity exists depends on the data that assign values to the accessible quantities, and the evidence that it exists depends on whether the measure determines a stable value for its target quantity from those values. Whether the measure is stable is a question put to the empirical world that only it can answer.

Still further, the theoretical justification cannot predict what the value is going to turn out to be. Any such prediction can only come from having obtained a value for nominally the same quantity from some other measure. Here again, therefore, only the empirical world can answer the question of what the value of the target quantity is. As a consequence, for a measure to yield a stable value—or, even better, a stable value amenable to increasing precision that is convergent with a value from another measure—in and of itself gives that value a *prima facie* claim to at least represent a physical magnitude in much the way that one's running into a table in the dark gives the perception of the contact a *prima facie* claim to involve an encounter with a physical object.

That the theoretical justification of a measure cannot dictate that the value of its target quantity will turn out to be stable, much less the value itself, is reminiscent of a requirement for measurements to "ground" a theory that van Fraassen adapted from Clark Glymour: "there must be an alternative possible outcome for the same measurements that would have refuted the hypothesis *on the basis of the same theoretically posited connections*."[58] The requirements of stability, convergence, and amenability to increasing precision in theory-mediated measurement parallel van Fraassen's requirements of "determinability" and "concordance" for grounding that he adapted from Herman Weyl. A measurement cannot fully

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ground a theory in van Fraassen's sense unless its measure is stable, convergent, and amenable to increasing precision.

He imposes a further requirement, however: "this determination can, may, and generally must be made *on the basis of theoretically posited connections*." Perrin's values for *W*, for example, could not have grounded molecular-kinetic theory without the assumption that the mean translational kinetic energies of the granules and the molecules of the liquid are the same. We do not disagree with van Fraassen about this. The contrast between him and us lies in our setting apart the measures in theory-mediated measurements and their relation to the data and data models from which they extract values in and of themselves, independently of their theoretical justification. And our point is that the measures and the values they yield *can* acquire special standing, a claim to permanence, in their own right that is independent of their theoretical justification.

Van Fraassen's term "grounding" is of course a metaphor, but one that we agree is appropriate for describing, for example, what the determinations of Avogadro's number examined in the body of this monograph did at the time for the molecular-kinetic theory of heat and the atomic-molecular theory of chemical composition. The question we have been considering in this section of our Postscript, by contrast, concerns why the values of theory-mediated measurement have sometimes persisted through change in theory. The corresponding metaphor for this, we submit, is that values which have acquired the sort of claim to permanence that we have proposed *anchor* theory change. They, and the measures yielding them (once suitably qualified), become constraints on theory change.

P.3.6. So What?

One might well ask why this should matter to anti-realists and realists disputing the scope and limits of theoretical knowledge. The reason does not lie in the values themselves, but in a further implication of evidence being able to eliminate virtually all possibility of their not corresponding to a physical magnitude. Evidence that does this is evidence that the measures yielding those values represent, at least to some precision over a circumscribed domain, physical regularities. Granted, these regularities tend to be highly local, and the measures representing them may turn out to be surrogates for representations that in some respect or other are preferable to them. Nevertheless, evidence supporting the permanence of a value obtained through a measure is evidence that the measure supports counterfactual claims. For example, the rate of rarefaction with height of granules in a dilute emulsion would be different, indeed in a specifiable way, if the value of *W* were different, say from a change in temperature, or if the granules

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of the emulsion were of a different buoyant weight. The relationship between the period of a small-arc circular pendulum and the length of its string would be different, again in a specifiable way, if the value for *g* were different, say from making the measurement at a different latitude, or if the size of the bob and hence the center of oscillation of the pendulum were different. And the pattern of departures from uniform angular motion, as viewed from the Earth, would be different in a specifiable way if the value of the eccentricity for Mars determined by Ptolemy's measure were more like that for Jupiter.

Several points need to be made about such counterfactuals. To begin with, they concern only quantities that enter into a measure, its so-called accessible quantities, manipulable or otherwise, and the value of its target quantity. Like the measures, they too therefore tend to be highly local. What supports them is the evidence that the measure represents a physical regularity at least to some precision over a circumscribed domain. We have argued earlier that evidence that the measure does this need focus only on the value of the target quantity, without having to take into consideration the physical interpretation assigned to it by the theoretical justification for the measure. But then, to the extent that the evidence for the stability and amenability to increasing precision of the measure is independent of its theoretical justification, and the evidence for its convergence with other measures is restricted to being so, the counterfactuals supported by the measure are also independent of its theoretical justification. The counterfactual about the rate of rarefaction of granules and their buoyant weight stated earlier, for example, holds independently of what *W* represents physically. Therefore, evidence that the value of the target quantity is going to persist through theory change is evidence as well for the counterfactuals supported by the measure. In other words, to the extent that this value constrains future theory change, so too do these counterfactuals.

This, however, entails that the counterfactuals be highly local in another respect. The counterfactuals concern alternative values of the quantities entering into the measures. The evidence supporting the measure, taken in isolation from its theoretical measure, involved variations of those values over a limited range. Consequently, the alternative values entering into the counterfactuals supported by the measure need to be restricted to the same range. If the alternative values are not so restricted, the counterfactual may well be invalidated by future theory change, for only the theoretical justification of a measure can license the range of its projectability. To give a historical example of this, not so far removed from theory-mediated measurement as it may at first appear, consider the counterfactuals pertaining to our solar system licensed by Newton's law of gravity. Before general relativity, they included counterfactuals in which the mass of the sun could be varied by orders of magnitude. The weak-field limit imposed on Newton's law by general relativity restricted the validity of the counterfactuals

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supported by it to a narrow range—or, in Glashow's phrasing, a narrow "envelope of applicability."

Narrow though they may be, the counterfactuals nevertheless identify which details in the world make a difference and, at least to some level of precision, the differences they make. The buoyant weight of the granules, for example, makes a difference to the rate of their rarefaction in Perrin's sedimentation equilibrium, but the viscosity of the fluid does not; and the size of the granules, but not their mass, makes a difference in their rate of diffusion in a dilute emulsion. We showed in Chapter 4 that the evidence Perrin developed for both of these claims was independent of molecular-kinetic theory, and hence that they hold independently of whether the mean translational kinetic energy of the granules is or is not the same as the mean translational kinetic energy of molecules in the fluid. We are now arguing that at least much of the evidence for them was also independent of whether W is the mean translational kinetic energy of the granules. But for just this reason, the details singled out in the counterfactuals as making a difference and the differences they make acquire a claim to permanence through theory change—whether as specified or under a re-description or a limitation of scope—when the value of the target quantity has such a claim.

The instrumentalism-realism debate has focused on overarching, usually fundamental theories. Laudan's once-accepted-subsequently-rejected examples involved mostly such theories too, challenging whether any evidence for them can ever assure that they will not be rejected in the future. If one were to consider with any care the question of what knowledge the sciences have given us over the last four centuries that Galileo did not have, surely the answer would not center on overarching fundamental theories. The best candidate for knowledge that the sciences have given us that Galileo could scarcely have imagined consists of literally millions of details in the world that make a difference and, at least to some level of precision, the differences they make. Correspondingly, the sense that we understand the world around us so much more thoroughly than those in Galileo's time derives far more from what the sciences have revealed about which details in the world make various differences, and which do not, than it does from their overarching, fundamental theories.

We are by no means denying that these theories have entered indispensably into establishing many of these details. Our principal point in this section of the Postscript is that whatever claim to permanence at least some such details have has often been far less dependent on overarching theory than is generally recognized. We submit, as a corollary to this point, that the extent to which theory has entered into claims to permanence, and precisely how it has, requires the sort of detailed analysis of the evidence illustrated in this monograph.

Nothing in this Postscript should be read as taking sides, one way or another, in the instrumentalist-realist debate. We have merely been putting forward some

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suggestions that might give the sides of the debate a more extensive common ground with one another and that might accordingly make it more tractable than it has tended to be. In this regard, we intend the Postscript to be an addendum to Howard Stein's more cited than heeded paper of 30 years ago, "Yes, but . . . Some Skeptical Remarks on Realism and Anti-Realism."[59] At the risk of oversimplifying what he said, we take his main claim to have been that close analysis of the history of the development of different areas of science will leave the two sides of the debate less at irreconcilable loggerheads with one another. We are suggesting that one promising topic for such analysis is the role of theory-mediated measurement in identifying which details in the world make a difference and what differences they make.

The twentieth century abounds with examples for which, at first glance, realists and anti-realists would seem to be diametrically opposed, but on further inspection might have trouble remaining so. We end with two more closely complementary examples than one might at first think of details revealed by theory-mediated measurements: (1) the angle between the two oxygen atoms in carbon dioxide is 180 degrees and hence the shape of the molecule is linear, in distinct contrast to other dioxides such as NO_2 (134.1 degrees) and SO_2 (119.5 degrees);[60] and (2) roughly 2,890 km below the surface of the earth, its interior changes from solid to liquid, and some further roughly 1,270 km below that, it changes from liquid to solid.[61] These both involve measured quantities to which we will almost surely never gain direct access. The listed values of the quantities were all obtained through representations that, in one way or another, simplified or idealized a more complicated state of affairs than was being taken into account. Nevertheless, all these quantities have claim to being permanent—the geophysical ones only once they are referenced to our present epoch insofar as the earth is gradually solidifying. We submit that examples like these can do more to clarify what, if anything, is ultimately under dispute between realists and anti-realists than the ether and molecule examples that have dominated the literature ever since Duhem.

Notes

1. Meyer (1899b), p. vi.
2. See Nernst (1969).
3. See Barkan (1999) for support for this claim.
4. That is, defenses of realism postdating Laudan (1981).
5. Duhem (2015); the French original was published in 1902.
6. Worrall (1989).

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7. We repeat here what Planck specifically said in the Preface to the second edition of his *The Theory of Heat Radiation*: "Meanwhile the experimental methods, improved in an admirable way by the labors of *E. Rutherford, E. Regener, J. Perrin, R. A. Millikan, The Svedberg*, and others, have without exception decided in favor of the value [of the elementary electric charge] deduced from the theory of radiation which lies between the values of *Perrin* and *Millikan*." Later in the book he derives a value of 4.67×10^{-10} esu from the value 1.34×10^{-18} ergs per degree Kelvin for k derived from blackbody radiation (along with values of 6.20×10^{23} for N_0 and 2.01×10^{-16} for α), and then adds: "Since absolute accuracy is claimed for the formulae here employed, the degree of approximation to which these numbers represent the corresponding physical constants depends only on the accuracy of the measurements of the two radiation constants, a and c_2" (Planck, 1991, p. 172f).
8. The adverb *exacté* occurs once in the first edition of the *Principia*: "I suspended the same weight of gold as exactly as I could in the center of oscillation of the other [pendulum]" (Book 3, Prop. 6). By contrast, *accuraté* has more than 80 occurrences (generally translated as "exactly"), and *quamproximé* (literally, *as nearest as possible*) more than 130.
9. Book 1, Proposition 3, Corollary 2.
10. For more on Newton's concern with the validity of inferences drawn from such claims, see Smith (2002).
11. Millikan's error, of course, was using a wrong value for viscosity. For details, see Birge (1945). Birge's revised value for e derived from the value for Avogadro's number and the Faraday constant was $4.8021 \pm 0.0006 \times 10^{-10}$ esu. The 2018 value, $4.80320471 \times 10^{-10}$ esu, readers should note, lies outside Birge's 1944 bands, though not so far outside as his 1944 value fell outside Millikan's bands. Estimates of error-bands often fall prey to undetected sources of systematic error, as illustrated by Millikan's 1917 bands.
12. See Newton's *Principia*, Book 1, Propositions 50–52 and the two corollaries of the latter. The formula Newton gives there has the imprecision arising from Huygens's assumption appearing in the seventh significant figure in the case of a uniformly dense earth.
13. Glashow (1997), p. 276.
14. In his doctoral dissertation, Dustin King (2018) has examined examples of both cross-checks and complementary evidence in his analysis of Cecilia Payne's groundbreaking determination of the extraordinarily high abundances of hydrogen and helium in stars. To calculate abundances at all, she had to take a substantial number of propositions to be true, some of which she was quite sure at the time were not true, at least as stated. Her dissertation, Payne (1925), provides a model for how to obviate the risks of leveraging results from propositions taken to be true for which the available evidence is at best limited and at worst, little different from wishful thinking.
15. Perrin (1990), p. 106f; Perrin (1912), p. 185f.
16. Perrin (1910), p. 73. Italics in the original.
17. See section 2.6 of Chapter 2 for a sketch of the struggle with the second law during the second half of the nineteenth century.

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18. To see why we have included Nernst's new heat theorem, consider the following remark by Planck in the Preface to the second edition of his *The Theory of Heat Radiation* (1991), p. vii:

    > For the hypothesis of quanta as well as the heat theorem of *Nernst* may be reduced to the simple proposition that the thermodynamics probability (Sec. 120) of a physical state is a definite integral number, or, what amounts to the same thing, that the entropy of a state has a quite definite positive value, which, as a minimum, becomes zero, while in contrast therewith the entropy may, according to the classical thermodynamics, decrease without limit to minus infinity. For the present, I would consider this proposition as the very quintessence of the hypothesis of quanta.

19. See Thomson and Thomson (1928), p. 264.
20. Ibid. Lorentz had recommended the correction to Kaufmann well before Einstein published his special relativity paper.
21. Nernst (1923), p. 327; (1921), p. 285 in the original German edition.
22. Thomson and Thomson (1928), p. 264.
23. Birge (1942), p. 127; and Mohr et al. (2016), p. 35009–35057.
24. The value Huygens listed for the distance of fall in the first second in his *Horologium Oscillatorium* of 1673 (Part 4, Prop. 26) was 15 Paris feet 1 inch, based on the length of 3 Paris feet, 8.5 lines (where a line is a twelfth of an inch) for a one-second pendulum (Huygens, 1986, p. 171); we have converted the latter value to a value for g in metric units. The 1958 value we have quoted is from Heiskanen and Vening Meinesz (1958), p. 85.
25. These values are taken from Neugebauer (1975), pp. 177ff. We have inverted Ptolemy's radius ratios, halved his double-eccentricities, and converted his hexadecimal values to decimal. Ptolemy's value for the eccentricity of Mars came from a third trial that has received some commentary; the value, in modern form, for the second trial was 0.098. G. W. Hill has shown why Ptolemy's value for the eccentricity of Mars was less accurate than his values for Jupiter and Saturn, namely an assumption that the eccentricities are small enough for him to have, in effect, dropped a term. See Hill (1900).
26. C. Wilson (1989), p. 180.
27. Fegley and Lodders (1998), p. 87f.
28. Laudan (1981).
29. Duhem (1991), pp. 7–9 and Part I, Chapter 3. Duhem concedes in these pages that the vibrational theory of sound explains, in his sense, acoustic phenomena, citing our being able to confirm the existence of the vibrations with our senses as distinguishing this case of the explanation of optical phenomena in terms of vibrations in the ether, the existence of which we are unable to confirm with our senses. In saying the latter, by the way, Duhem does not reject the electromagnetic wave theory of light, taken to consist of measurable oscillations of the strength of the local electric and magnetic potentials in the electromagnetic field. See his (2015).
30. This point and its implications with respect to the realism-instrumentalism debate is stressed by Jody Azzouni in his (2004) and in his paper coauthored with Otávio Bueno (Azzouni and Bueno, 2016). While these papers do not focus on theory-mediated

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measurement in developing the point, our doing so has been significantly influenced by them and by remarks Azzouni made, a propos of it, in a seminar on realism in which our monograph was a central element.

31. See Perrin (1910), pp. 34–40.
32. Ibid., p. 32f.
33. van Fraassen (2008), pp. 166ff; Suppes (1962). As we acknowledged in Chapter 1, we have elected to ignore the philosophic issues arising from the gaps that almost always occur between what are usually called "raw data" and the statements of the numerical results of experiments and measurements in the ensuing publications, instead granting the authority of the latter. This was in keeping first with our endeavor to view the topics of this book as much as possible as someone writing a review article at the time would have, and second with our focus on the theoretical relations—that is, the "measures"—bridging the gap between parameters that are physically measured and target parameters. For more recent investigations of the representation of data than those of Suppes, we recommend Tal (2012).
34. Suppes (1962), p. 258.
35. For more details on the seconds-pendulum measure and why its precision came to be so important, see Schliesser and Smith (1996).
36. See Heiskanen and Meinesz (1958), pp. 84–101.
37. See Chang (2004), especially pp. 49–56 on the first claim and pp. 62–64 on the second. For original sources, see Cavendish et al. (1777), on the first claim; and de Luc (1772), on the second.
38. Duhem (1991), Part I, Chapter 4.
39. See section 2.1 in Chapter 2 for a summary of Ostwald's use of atoms and molecules in his expositions of phenomena in physical chemistry, in spite of his constant reminders that they are purely hypothetical.
40. Weinberg (2001), p. 198. Weinberg terms the first part "hard"—in the sense of durable—and the second "soft," adding of the latter that "it does change; we no longer believe in Maxwell's ether, and we know that there is more to nature than Newton's particles and forces." He then adds,

    > The changes in the soft part of scientific theories also produce changes in our understanding of the conditions under which the hard part is a good approximation. But after our theories reach their mature forms, their hard parts represent permanent accomplishments. If you have bought one of those T-shirts with Maxwell's equations on the front, you may have to worry about its going out of style, but not about its becoming false. We will go on teaching Maxwellian electrodynamics as long as there are scientists. I can't see any sense in which the increase in scope and accuracy of the hard parts of our theories is *not* a cumulative approach to truth.

    In a subsequent response to a challenge to these remarks, he added (p. 208),

    > But approximate theories are not merely approximately true. They can make a statement that, though it refers to an approximation, is nevertheless precisely true. For instance, although Maxwell's equations give only an approximate account of electric and magnetic fields, it is precisely true that the error introduced by using Maxwell's equations to calculate these fields can be made as

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> small as one likes by considering fields that are sufficiently weak and slowly varying. This is part of the reason that Maxwell's equations are a permanent part of physical science.

We acknowledge that the points made here by Weinberg and conversations Smith has had with him have influenced the way we put a number of things in this monograph, especially in this Postscript.

41. Perrin (1910), p. 46.
42. See Huygens (1986), especially pp. 69ff and 170–175; and Yoder (1988), pp. 16–33.
43. Azzouni (2002), especially p. 2 and pp. 63–70; and (2010), pp. 187–201.
44. For the case of the radius ratios, see Swerdlow (2004).
45. See section 4.3 of Chapter 4.
46. See, for example, Perrin's response to such a challenge in his (1908b).
47. See section 4.2 of Chapter 4 for how the mean translational kinetic energy enters into the first measure, and section 4.4 for how it enters into the second.
48. We include the greater-compatibility possibility to allow for advances in theory to give grounds for replacing the physical interpretation of *W* with one for which the mean translational kinetic energy of the granules at a given temperature comes to be viewed as a surrogate under certain qualifications.
49. Newton's "Moon test," readers should note, involved a convergence between Huygens's value for the acceleration of gravity at the surface of the Earth and the inverse-square acceleration of the moon toward the Earth inferred from its orbital motion.
50. See Huygens (1986), p. 168ff, where he discusses not only the need to measure the length to the center of oscillation to achieve greater precision, but some other moves as well.
51. Perrin (1908d), p. 595. Perrin, of course, made this claim about the measure for Avogadro's number obtained from the one we are considering by assuming that his value for *W* is the same as the value for the mean translational kinetic energy of molecules at the same temperature; that this assumption turned out to be false does not entail that the corresponding claim about *W* was false.
52. Perrin (1990), p. 124.
53. Planck (1991), p. 168.
54. Ibid., p. 173.
55. Birge (1945).
56. See, for example, Jeans (1916), p. 8.
57. Here we are very much echoing the corresponding parallel Azzouni put forward in his (2004), though the point we are making with it is much narrower than his, since here too ours is confined to the results of measurement.
58. van Fraassen (2009), p. 12. Italics in the original.
59. Stein (1989).
60. See Bernath (2016), especially pp. 262–274. For some early history of this discovery, see Dennison (1931).
61. See Masters and Shearer (1995). For some early history of this discovery, see Brush (1996), pp. 184–202.

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